

Def: (group action on X) $G \xrightarrow{a} \text{Aut}(X)$, $X \in \mathcal{C}$.

- for $\forall g \in G$, we have an morphism $ag): X \rightarrow X$
- for $\forall g, h \in G$, $a(g \cdot h) = ag) \circ ah)$
- $a(e) = \text{id}_X$, $e \in G$.

Prop:

Let R be a commutative ring and G a group. The following are equivalent data on an R -module M :

- (1) a monoid homo $G \rightarrow \text{End}_R(M)$
- (2) a group homo $G \rightarrow \text{Aut}_R(M)$
- (3) a homo of R -algebras $R[G] \rightarrow \text{Hom}_R(M, M)$
- (4) an $R[G]$ -module str on M whose underlying R -module is M .

Proof:

(1) $\Leftrightarrow$ (2)

$\rho: G \rightarrow \text{End}_R(M)$ is a monoid homo. Since G is a group, for $\forall g \in G$,
 $\rho(g)\rho(g^{-1}) = \rho(gg^{-1}) = \rho(e) = \text{id}_M$

Similarly, $\rho(g^{-1})\rho(g) = \text{id}_M$. So $\rho(g)$ is invertible. So $\rho(g) \in \text{Aut}_R(M)$. So

ρ is a group homo $G \rightarrow \text{Aut}_R(M)$.

(1) $\Leftrightarrow$ (3)

$\rho: G \rightarrow \text{End}_R(M)$. Define an R -linear map $\tilde{\rho}: R[G] \rightarrow \text{End}_R(M)$ by

$$\tilde{\rho}\left(\sum_{g \in G} r_g g\right) = \sum_{g \in G} r_g \rho(g).$$

Easy to see $\tilde{\rho}$ is an R -algebra homo.

Conversely, for an R -algebra homo $\varphi: R[G] \rightarrow \text{End}_R(M)$, define $\rho(g) = \varphi(g)$.

It is easy to see $\rho: G \rightarrow \text{End}_R(M)$ is a monoid homo.

(3) $\Leftrightarrow$ (4)

$\varphi: R[G] \rightarrow \text{End}_R(M)$ is an R -algebra homo. Define an $R[G]$ -action on M by $a \cdot m := \varphi(a)(m)$, $a \in R[G]$, $m \in M$. So the underlying R -module is exactly the given M .

Conversely, suppose M has an $R[G]$ -module str whose underlying R -module is M . Define $\varphi: R[G] \rightarrow \text{End}_R(M)$ by $\varphi(a)(m) = a \cdot m$. So φ is an R -algebra homo. $\square$

Def:

A rep of G over a commutative ring R is an $R[G]$ -module.

Exp: (sign rep)

$G = C_2 = \langle \tau \rangle$. $1 \neq -1$ in R . The sign rep of G is the 1-dim rep in which τ acts as -1 .

Exp:

$\rho_G = R[G]$, the regular rep. $1 = R^{\oplus n}$, trivial rep.

ρ_G has a 1-dim trivial subrep generated by $\sum_{g \in G} g$.

Def: (irreducible)

V is an irr rep iff the only subrep of V are 0 and V .

Thm. (Maske)

Suppose that k is a field of characteristic not dividing $|G|$. Then, every rep splits as a direct sum of irr reps.

Sketch of proof:

We just need to show for subrep $V \subset W$, $\exists$ a subrep $U \subset W$, s.t. $V \oplus U \cong W$. Let $p: W \rightarrow W$ be any linear projection onto the subspace V . Define $\varphi: W \rightarrow W$, $\varphi(w) = \frac{1}{|G|} \sum_{g \in G} g \cdot p(g^{-1} \cdot w)$. We can know φ is G -equivariant and $\varphi \circ \varphi = \varphi$. So let $U = \ker(\varphi)$, $W \cong V \oplus U$. $\square$

Ex:

$$\rho_{C_2} \cong 1 \oplus 1_{\text{sgn}}, \quad \rho_{C_3} \cong 1 \oplus \lambda_3, \quad \rho_{C_5} \cong 1 \oplus \lambda_5 \oplus \lambda_{5,2}$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \frac{2}{3}\pi & \frac{2}{5}\pi & \frac{4}{5}\pi \end{array}$$

Def: (real rep ring)

For a finite group G , the real rep ring of G , denoted $RO(G)$, is the quotient

$$RO(G) = \mathbb{Z} \{ \text{iso classes of finite-dim real } G\text{-reps} \} / \langle [V \oplus W] - [V] - [W] \rangle$$

$$\cong \bigoplus_{\text{brrep}(G)} \mathbb{Z}$$

$$[V] + [W] := [V \oplus W]$$

$$[V] \cdot [W] := [V \otimes_{\mathbb{R}} W]$$

$$g \cdot (v \otimes w) = (g \cdot v) \otimes (g \cdot w)$$

Ex:

$$G = C_2. \text{RO}(C_2) \cong \mathbb{Z}\{1\} \oplus \mathbb{Z}\{\text{sgn}\}$$

$$(\text{sgn})^2 = \text{sgn} \otimes \text{sgn} = 1, \text{RO}(C_2) \cong \mathbb{Z}[x]/x^2 - 1$$

$$G = C_3. \text{RO}(C_3) \cong \mathbb{Z}\{1\} \oplus \mathbb{Z}\{\lambda_3\}$$

$$[\lambda_3]^2 = 2 + [\lambda_3], \text{RO}(C_3) \cong \mathbb{Z}[\lambda]/(\lambda^2 - \lambda - 2)$$

Def (pullback)

Let $\varphi: H \rightarrow G$ be a group homo. Define the pullback $\varphi^*(V)$ of a G -rep along φ to be the same underlying vector space and with action given by $H \xrightarrow{\varphi} G \xrightarrow{\alpha} \text{End}(V)$.

Def (restriction)

Let V be a G -rep, and $i: H \hookrightarrow G$. Define the restriction of V to H , denote $\downarrow_H^G(V)$ or $\text{Res}_H^G V$, as the pullback $i^*(V)$.

Ex

$$G = C_4 = \langle \sigma \rangle, H = \langle \sigma^2 \rangle \leq C_4.$$

$$\text{RO}(C_4) = \mathbb{Z}\{1, \text{sgn}, \lambda_4\} \rightarrow \text{RO}(C_2) = \mathbb{Z}\{1, \text{sgn}\}$$

$$1 \mapsto 1, \text{sgn} \mapsto 1, \lambda_4 \mapsto 2 \cdot \text{sgn}$$

Def (induced rep)

$H \leq G$. V is an H -rep. Define the induced rep of V up to G , denoted $\uparrow_H^G(V)$ or $\text{Ind}_H^G V$, as the tensor product $k[G] \otimes_{k[H]} V$.

$$\uparrow_H^G(V_1 \oplus V_2) \cong \uparrow_H^G(V_1) \oplus \uparrow_H^G(V_2).$$

$\uparrow_H^G: \text{RO}(H) \rightarrow \text{RO}(G)$ is a group homo but not a ring homo.

$T \subset G$ is a set of represents of left cosets G/H . $G = \bigcup_{t \in T} tH$

$$k[G] \otimes_{k[H]} V \cong \left(\bigoplus_{t \in T} t k[H] \right) \otimes_{k[H]} V \cong \bigoplus_{t \in T} (t k[H] \otimes_{k[H]} V) \cong \bigoplus_{t \in T} V.$$

The way G acts on $\bigoplus_{t \in T} V$: for $g \in G$,

$$tH \mapsto gtH, \text{ let } gt = t'h, t' \in T, h \in H,$$

$$g \cdot (t \otimes v) = t' \otimes hv. \rightarrow \text{acts on corresponding } V.$$

permutation of summands

$$\dots \oplus_{t'} V \oplus \dots \oplus_t V \oplus \dots$$

Prop
If $H \leq K \leq G$, then for $\forall H$ -rep V , we have

$$\uparrow_H^G(V) \cong \uparrow_K^G(\uparrow_H^K(V))$$

Ex
 $\rho_G = \text{Res}_G^G \cong \text{Res}_G^G \otimes_{\mathbb{R}} \mathbb{R} = \uparrow_e^G(1).$

For $H \leq G$, $\uparrow_H^G(\rho_H) = \rho_G.$

$C_2 \leq C_4$, $\uparrow_{C_2}^{C_4}(1) = \text{Res}_{C_2}^{C_4} \otimes_{\mathbb{R}[C_2]} 1 \cong \text{Res}_{C_2}^{C_4/C_2}.$

Prop
Let $H \leq G$. Inclusion is left adjoint to restriction

$$\uparrow_H^G: \text{Mod}_{\mathbb{R}[H]} \rightleftarrows \text{Mod}_{\mathbb{R}[G]}: \downarrow_H^G$$

$$\text{Hom}_{\mathbb{R}[G]}(\mathbb{R}[G] \otimes_{\mathbb{R}[H]} V, W) \cong \text{Hom}_{\mathbb{R}[H]}(V, \downarrow_H^G W)$$

Ex:
 $\downarrow_H^G: (f: \rho_G \rightarrow V) \mapsto (k \rightarrow \downarrow_e^G(V))$

Prop (projection formula)

$H \leq G$. Then $\uparrow_H^G(-): RO(H) \rightarrow RO(G)$ is an $RO(G)$ -module homo.

This means for $V \in RO(G)$ and $W \in RO(H)$, we have an iso

$$\uparrow_H^G(\downarrow_H^G(V) \otimes W) \cong V \otimes \uparrow_H^G(W).$$

$\downarrow_H^G: RO(G) \rightarrow RO(H)$ is a ring homo. So $RO(H)$ can be considered as an $RO(G)$ -module by $V \cdot W := \downarrow_H^G(V) \otimes W$, $V \in RO(G)$, $W \in RO(H)$. If

$\uparrow_H^G(-)$ is an $RO(G)$ -module, then

$$\uparrow_H^G(\downarrow_H^G(V) \otimes W) = \uparrow_H^G(V \cdot W) = V \cdot \uparrow_H^G(W) = V \otimes \uparrow_H^G(W).$$

exp

$$\uparrow_{C_2}^{C_4}(\sigma) = \lambda_4. \quad \lambda_4 \otimes \uparrow_{C_2}^{C_4}(\sigma) = \uparrow_{C_2}^{C_4}(\downarrow_{C_2}^{C_4}(\lambda_4) \otimes \sigma).$$

$$\text{So } \lambda_4 \otimes \lambda_4 = \uparrow_{C_2}^{C_4}(2\sigma \otimes \sigma) = \uparrow_{C_2}^{C_4}(2) = 2 \oplus 2\sigma$$

Def (coinduction)

$H \leq G$. Define coinduction $\text{Coind}_H^G: \text{Mod}_{k[H]} \rightarrow \text{Mod}_{k[G]}$ by

$$\text{Coind}_H^G(V) = \text{Hom}_{k[H]}(k[G], V)$$

G acts as follows

$$g \cdot f(\sum_{t \in G} c_t t) = f(\sum_{t \in G} c_t t \cdot g)$$

Prop:

$H \leq G$. Coinduction is right adjoint to restriction.

$$\downarrow_H^G: \text{Mod}_{k[G]} \rightleftarrows \text{Mod}_{k[H]}: \text{Coind}_H^G$$

$$\text{Hom}_{k[G]}(M, \text{Coind}_H^G V) \cong \text{Hom}_{k[H]}(\downarrow_H^G M, V)$$

Prop:

$H \leq G$. There is a natural iso of functors $\uparrow_H^G(-) \cong \text{Coind}_H^G$.

This gives

$$k[G/H] = \uparrow_H^G(1) \cong \text{Coind}_H^G(1) \cong \text{Hom}_{k[H]}(k[G], 1) \cong \text{Hom}_k(k[H \backslash G], 1)$$

Def (double coset)

$H, K \leq G$. We can think H acts on the left of G and K acts on the right of G . The quotient respect to both of these actions is the set $H \backslash G / K$ of double cosets of H and K in G .

$$HgK = \{x \in G \mid x = h g k \text{ for some } h \in H \text{ and } k \in K\}$$

$$K^g := g K g^{-1}$$

$$c_g(K) := K^g = g K g^{-1}$$

Thm (Mackey's double coset formula)

For subgroups H and K of G and an H -rep V , we have a decomposition of K -reps

$$\downarrow_K \uparrow_H^G(V) \cong \sum_{HgK \in H \backslash G / K} \uparrow_{H^g \cap K}^K (c_g^*(\downarrow_{H \cap K^g}^H(V)))$$

$$R\mathcal{O}(H) \xrightarrow{\text{Res}} R\mathcal{O}(H \cap K^g) \xrightarrow[c_g^*]{\cong} R\mathcal{O}(H^{g^{-1}} \cap K) \xrightarrow{\text{Ind}} R\mathcal{O}(K)$$

$$R\mathcal{O}(H) \xrightarrow{\text{Res}} R\mathcal{O}(G) \xrightarrow{\text{Ind}} R\mathcal{O}(K)$$

Cor

Suppose that G is abelian and $H=K$. Then the composition

$$R\mathcal{O}(H) \xrightarrow{\uparrow_H^G} R\mathcal{O}(G) \xrightarrow{\downarrow_H^G} R\mathcal{O}(H)$$

is multiplication by the index $|G/H|$ of H in G .

Ex

$$C_2 \leq C_4 \quad RO(C_2) \xrightarrow{\uparrow_{C_2}^{C_4}} RO(C_4) \xrightarrow{\downarrow_{C_2}^{C_4}} RO(C_2)$$

$$1 \mapsto 1 \oplus \sigma \mapsto 2$$

$$\sigma \mapsto \lambda_4 \mapsto 2\sigma$$

Def (fixed points)

Given a G -rep, or more generally an $R[G]$ -module M for some commutative ring R , we define the fixed points

$$M^G = \{m \in M \mid g \cdot m = m \text{ for all } g \in G\}$$

Prop

The fixed points functor is right adjoint to the trivial G -action functor

$$\text{triv} = \eta^*: \text{Mod}_R \rightleftarrows \text{Mod}_{R[G]} : (-)^G$$

Def (H -fixed points)

Given a G -rep, or more generally an $R[G]$ -module M for some commutative ring R , define the H -fixed points functor as the composite

$$\text{Mod}_{R[G]} \xrightarrow{\downarrow_{H^G}^G} \text{Mod}_{R[H]} \xrightarrow{(-)^H} \text{Mod}_R$$

Ex

$G = K_4$. The L -fixed points homo is $1 \mapsto 1, p_1^*(\sigma) \mapsto 0, m^*(\sigma) \mapsto 0, p_2^*(\sigma) \mapsto 1$.

Def (Mackey functor)

A Mackey functor $\underline{M}$ for the group G consists of the following data

- an abelian group $\underline{M}(H)$ for each subgroup $H \leq G$
- a restriction homo $R_H^K: \underline{M}(K) \rightarrow \underline{M}(H)$ where $H \leq K$
- a transfer or induction homo $I_H^K: \underline{M}(H) \rightarrow \underline{M}(K)$, where $H \leq K$
- a conjugation homo $c_g: \underline{M}(H) \rightarrow \underline{M}(H^g)$ for each $H \leq G$ and $g \in G$.

Exp

For G , $\underline{RO}_G(H) := RO(H)$. So $\underline{RO}_G$ is a Mackey functor. The conjugation by g is an iso $H \rightarrow H^g$. So $(c_g)^*: RO(H^g) \rightarrow RO(H)$. We need $c_g: RO(H) \rightarrow RO(H^g)$.
So $c_g := (c_{g^{-1}})^*$.

Def

M is an $R[G]$ -module. We define a Mackey functor $\underline{F}(M)$ by $\underline{F}(M)(H) = M^H$, $H \leq G$. If $H \leq K$, then $R_H^K: M^K \rightarrow M^H$ is the inclusion. $I_H^K: M^H \rightarrow M^K$ is given by

$$I_H^K(m) = \sum_{kH \in K/H} k \cdot m$$

$c_g: M^H \rightarrow M^{H^g}$ is given by acting by the element g .

Exp:

M is a trivial G -module. A is an abelian group with trivial

A -action, the resulting Mackey functor is the constant Mackey functor

$\underline{A}$ at A . All R_H^K and c_g are id. $I_H^K: \underline{A}(H) \rightarrow \underline{A}(K)$ is multiplication by the index of $|K/A|$.

$M = \mathbb{Z}[C_2]$.

diagonal map $\downarrow$
 $\mathbb{Z} \xrightarrow{\Delta} \mathbb{Z}[C_2] \xleftarrow{\nabla} \mathbb{Z}$
 fold map $\nwarrow$
 $\mathbb{Z}[C_2] \xrightarrow{\quad} \mathbb{Z}[C_2]$
 Lewis diagram

Def

A map of G -Mackey functors $\underline{M} \rightarrow \underline{N}$ is a collection of homs $\underline{M}(H) \rightarrow \underline{N}(H)$, as H runs over subgroups of G , which are compatible with the restriction, transfer and conjugation homs. Then we get a category of Mackey functors for G , denoted by $\text{Mack}(G)$.

Evaluation at the trivial subgroup gives

$$ev_e: \text{Mack}(G) \rightarrow \text{Mod } \mathbb{Z}[G]$$

$$\text{for } \underline{M} \in \text{Mack}(G), ev_e(\underline{M}) = \underline{M}(e).$$

Prop

For each G , the evaluation functor ev_e is left adjoint to the fixed point functor

$$ev_e: \text{Mack}(G) \rightleftarrows \text{Mod } \mathbb{Z}[G] : F$$

Def

M is an $R[G]$ -module. Define a Mackey functor $\underline{Q}(M)$ by $\underline{Q}(M)(H) = M/H$.

For $H \leq K$, $R_H^K: M/K \rightarrow M/H$ is given by

$$R_H^K(\bar{m}) = \sum_{kH \in K/H} \bar{k} \cdot \bar{m}$$

$I_H^K: M/H \rightarrow M/K$ is the quotient map. $C_g: M/H \rightarrow M/H^g$ is given

$$\text{by } C_g(\bar{m}) = \overline{g \cdot m}.$$

Def

A is an abelian group, equipped with a trivial action of G . Then

$Q(A)$ is the dual constant Mackey functor, denoted A^* . The transfer and conjugation are id. The restrictions are multiplication by the index of H in K .

Prop

The evaluation functor is right adjoint to the quotient functor.

$$Q: \text{Mod}_{\mathbb{Z}[G]} \rightleftarrows \text{Mack}(G): \text{ev}_e$$

$$\text{Hom}_{\text{Mack}(G)}(Q(A), \underline{M}) \cong \text{Hom}_{\mathbb{Z}[G]}(A, \underline{M}(e))$$

Def (Burnside ring)

G is a finite group. The Burnside ring of G , denoted $A(G)$, is the quotient

$$A(G) = \mathbb{Z}\{\text{iso classes of finite } G\text{-sets}\} / \langle [X \amalg Y] - [X] - [Y] \rangle$$

$$A(G) \cong \bigoplus_{\text{conj}(G)} \mathbb{Z}$$

$$[X] + [Y] = [X \amalg Y]$$

$$[X] \cdot [Y] = [X \times Y]$$

$$g \cdot (x, y) = (gx, gy)$$

This is because for any finite

G -set X , $X \cong \amalg_i G/H_i$, where $H_i \leq G$ is a stabilizer for some point.

$X \cong \amalg_{\text{orbits } [x_i]} G/H_{x_i}$ $\hookrightarrow$ each i corresponds to one orbit each rep point in each orbit.

Def

For any G , let $A_G(H)$ be a group $A(H)$. $R_H^K: A(K) \rightarrow A(H)$ takes a set

X with an action of K and restricts the action to the subgroup H .

$I_H^K: A(H) \rightarrow A(K)$ takes a set X with an action of H and produces the

K -set $K \times_H X$. $C_g: A(H) \rightarrow A(H^g)$ takes $H \rightarrow \text{Aut}(X)$ to $H^g \xrightarrow{C_{g^{-1}}} H \rightarrow \text{Aut}(X)$.

Prop The evaluation functor ev_G is right adjoint to the tensoring with $\underline{A}_G$

$$-\otimes \underline{A}_G: Mod \mathbb{Z} \rightleftarrows Mack(G): ev_G$$

$$Hom_{Mack(G)}(B \otimes \underline{A}_G, \underline{M}) \cong Hom_{\mathbb{Z}}(B, \underline{M}(G))$$

$$\mathbb{Z} \otimes \underline{A}_G \cong \underline{A}_G \rightarrow \underline{M} \leftrightarrow \text{Maps of } \mathbb{Z}\text{-modules } \mathbb{Z} \rightarrow G \\ \leftrightarrow \text{elements of } \underline{M}(G)$$

Prop For any group G , the Mackey functors $\mathcal{Q}[\mathbb{Z}[G]]$ and $\underline{A}_G$ are projective Mackey functors.

Def Let $\underline{M} \in Mack(G)$ and $H \leq G$. Define the restriction $\downarrow_H^G \underline{M} \in Mack(H)$ by $\downarrow_H^G \underline{M}(K) = \underline{M}(K)$, $K \leq H$. The restriction, transfer, and conjugation are those of the G -Mackey functor $\underline{M}$.

Def Let $\underline{M} \in Mack(H)$, thought of as indexed over finite H -sets, and let $H \leq G$. Define the induction $\uparrow_H^G \underline{M}$ of $\underline{M}$ from H to G by setting

$$\uparrow_H^G \underline{M}(X) = \underline{M}(\downarrow_H^G X)$$

where here we denote by $\downarrow_H^G X$ the restriction of G -action on X down to the H -action. The restriction, transfer, and conjugation are also given by this formula.

Actually, $\underline{M}(K) := \underline{M}(G/K)$. And for every finite G -set X , $X \cong \coprod_i G/K_i$.

So we can define $\underline{M}(X) := \bigoplus_i \underline{M}(K_i)$.

So Mackey functor can be defined as an additive functor starting from Burnside category. $\underline{M}: \mathcal{B}_G \rightarrow \mathcal{A}b$. So we can give

$$\uparrow_H^G \underline{M}(K) = \bigoplus_{HgK \in H \backslash G/K} \underline{M}(H \cap K^g)$$

As definition, $(\uparrow_H^G \underline{M})(G/K) = \underline{M}(\downarrow_H^G(G/K))$. So we need to consider orbits in G/K as an H -set. For a represent $gK \in G/K$, whose H -stabilizer is $HgK = \{h \in H \mid hgK = gK\}$. $hgK = gK$ means $g^{-1}hg \in K$, which is $h \in H \cap gKg^{-1}$.

So $HgK = H \cap K^g$. So this orbit is $H/(H \cap K^g)$. So

$$\downarrow_H^G(G/K) \cong \coprod_{HgK \in H \backslash G/K} H/(H \cap K^g)$$

Then,

$$\begin{aligned} \uparrow_H^G \underline{M}(K) &= \underline{M}(\downarrow_H^G(G/K)) = \bigoplus_{HgK \in H \backslash G/K} \underline{M}(H/(H \cap K^g)) \\ &= \bigoplus_{HgK \in H \backslash G/K} \underline{M}(H \cap K^g) \end{aligned}$$

Prop

$H \leq G$. The restriction functor $\downarrow_H^G(-): \text{Mack}(G) \rightarrow \text{Mack}(H)$ is both left and right adjoint to the induction functor $\uparrow_H^G(-): \text{Mack}(H) \rightarrow \text{Mack}(G)$.

Prop

$H \leq G$. ev_H is right adjoint to tensoring with $\uparrow_H^G \underline{A}_H$

$$\begin{array}{ccc} \text{Mod } \mathbb{Z} & \xrightarrow{- \otimes \underline{A}_H} & \text{Mack}(H) \\ \uparrow \text{ev}_H & & \downarrow \downarrow_H^G \\ \text{Mack}(H) & \xrightarrow{\uparrow_H^G} & \text{Mack}(G) \end{array}$$

(with a curved arrow from $\text{Mod } \mathbb{Z}$ to $\text{Mack}(G)$ labeled $- \otimes \uparrow_H^G \underline{A}_H$)

Prop

$H \leq G$. The G -Mackey functor $\uparrow_H^G A_H$ is a projective Mackey functor.

Cor

The abelian category $\text{Mack}(G)$ has enough projectives. That means for any $\underline{M} \in \text{Mack}(G)$, $\exists$ a projective Mackey functor $\underline{P} \in \text{Mack}(G)$ s.t. $\underline{P} \twoheadrightarrow \underline{M}$.

Def

Let $N \trianglelefteq G$ and $Q = G/N$, and $q: G \rightarrow Q$ for the quotient map. For $\forall \underline{M} \in \text{Mack}(Q)$, we define $\text{Inf}_Q^G \underline{M}$, the inflation of $\underline{M}$ along q , by

$$\text{Inf}_Q^G \underline{M}(H) = \begin{cases} \underline{M}(H/N) & N \leq H \\ 0 & \text{else} \end{cases}$$

Prop

$\text{Inf}_e^{C_2}: \text{AbGrp} \rightarrow \text{Mack}(C_2)$ has left and right adjoints. The left adjoint is

$$\underline{M} \mapsto \underline{M}(C_2) / \text{im } I_e^{C_2}$$

and the right adjoint is

$$\underline{M} \mapsto \ker R_e^{C_2}: \underline{M}(C_2) \rightarrow \underline{M}(e)$$

Def

Given a $\mathbb{Z}[G]$ -module M , the coinvariants of M is the abelian group

$$M/G = \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$$

where the ring homo $\mathbb{Z}[G] \rightarrow \mathbb{Z}$ is the map sending each generator g to 1. Similarly, for a G -rep V , the coinvariant is the vector space $V_G = k \otimes_{k[G]} V$.

This is the quotient of V by the vector space spanned by elements of the form $v - g.v$.

Def

Define the group homology of G to be the groups

$$H_n(G) = \text{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, \mathbb{Z})$$

And define the homology of G with coefficients in M ($\mathbb{Z}$ -module) to be

$$H_n(G; M) = \text{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M).$$

And $H_0(G; M) \cong M/G$.

Prop

A is an abelian group, or more generally a G -module, in which multiplication by order $|G|$ is an iso. Then, the higher homology groups $H_n(G; A)$ ($n > 0$) all vanish.

Lemma

$S \rightarrow T$ is a flat ring extension, meaning that T is flat as an S -module. Then for any S -module M and T -module N , we have an iso

$$\text{Tor}_n^S(M, N) \cong \text{Tor}_n^T(M \otimes_S T, N).$$

Cor

For any finite group G , the higher homology groups with coefficients in $\mathbb{Q}$ or $\mathbb{R}$ vanish.

Def

Let G be a group. Define the group cohomology of G to be the groups

$$H^n(G) = \text{Ext}_{\mathbb{Z}[G]}^n(\mathbb{Z}, \mathbb{Z})$$

For any $\mathbb{Z}[G]$ -module M , define the cohomology of G with coefficients in M

$$H^n(G; M) = \text{Ext}_{\mathbb{Z}[G]}^n(\mathbb{Z}, M)$$

Prop

We have

$$H^0(G; M) = \text{Ext}_{\mathbb{Z}[G]}^0(\mathbb{Z}, M) \cong M^G$$

Def:

G and K are groups. R is any commutative ring. P_* is a free resolution of $\mathbb{Z}$ over G and Q_* is a free resolution over K . Define the cross product $\times$ to be the composition

$$\begin{array}{ccc}
 [\alpha] \otimes [\beta] & \xrightarrow{\quad} & [\alpha \otimes \beta] \\
 H^n(G; R) \otimes H^l(K; R) & \xrightarrow{\quad} & H^{n+l}(\text{Hom}_G(P_*, R) \otimes \text{Hom}_K(Q_*, R)) \\
 & \searrow \times & \downarrow \\
 & & H^{n+l}(\text{Hom}_{G \times K}(P \otimes Q)_*, R) \\
 & & \parallel \\
 & & H^{n+l}(G \times K; R) \\
 & & \uparrow \\
 & & [\bar{\Phi}(\alpha \otimes \beta)]
 \end{array}$$

where $\bar{\Phi}(\alpha \otimes \beta)(p \otimes q) = \alpha(p) \beta(q)$

Def

G is a group and R is a commutative ring. Denote by $\Delta: G \rightarrow G \times G$ the diagonal homo. Define the cup products as the composition

$$\begin{array}{ccc} H^n(G; R) \otimes H^l(G; R) & \xrightarrow{\cup} & H^{n+l}(G \times G; R) \\ & \searrow \checkmark & \downarrow \Delta^* \\ & & H^{n+l}(G; R) \end{array}$$

Exp

$G = C_2$. Denote $\{e, g\} := C_2$.

$$P_*: \dots \xrightarrow{\cdot(e-g)} \overset{P_2}{\mathbb{Z}[C_2]} \xrightarrow{\cdot(e+g)} \overset{P_1}{\mathbb{Z}[C_2]} \xrightarrow{\cdot(e-g)} \overset{P_0}{\mathbb{Z}[C_2]} \rightarrow \mathbb{Z}$$

$$\begin{array}{ccccc} \dots & \xrightarrow{0} & \mathbb{Z} & \xrightarrow{\cdot 2} & \mathbb{Z} & \xrightarrow{0} & \mathbb{Z} \\ & & \parallel & & \parallel & & \parallel \\ & & P_2 \otimes \mathbb{Z} & & P_1 \otimes \mathbb{Z} & & P_0 \otimes \mathbb{Z} \\ & & \mathbb{Z}[C_2] & & \mathbb{Z}[C_2] & & \mathbb{Z}[C_2] \end{array}$$

So

$$H_n(C_2) = \begin{cases} \mathbb{Z} & n=0 \\ \mathbb{Z}/2 & n \text{ odd} \\ 0 & n \text{ even } n > 0 \end{cases}$$

We know

$$H^n(C_2) = \begin{cases} \mathbb{Z} & n=0 \\ \mathbb{Z}/2 & n \text{ even}, n > 0 \\ 0 & n \text{ odd} \end{cases}$$

Let $x \in H^2(C_2) \cong \mathbb{Z}/2$ be a generator. x is represented by the C_2 -equiv

homomorphism $f_x: P_2 = \mathbb{Z}[C_2] \rightarrow \mathbb{Z}, 1 \mapsto 1$. $x \cdot x \in H^4(C_2)$ is given by

$$(P \otimes P)_4 \twoheadrightarrow P_2 \otimes P_2 \xrightarrow{f_x \otimes f_x} \mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}.$$

$$\begin{array}{ccc}
 P_4 = \mathbb{Z}[C_2] & \xrightarrow{(1 \otimes 1) \quad g \otimes g \quad -1 \otimes 1 \quad -g \otimes g \quad 1 \otimes 1} & (P_0 \otimes P_4) \oplus (P_1 \otimes P_3) \oplus (P_2 \otimes P_2) \oplus (P_3 \otimes P_1) \oplus (P_4 \otimes P_0) \\
 \downarrow 1+g & & \downarrow \begin{bmatrix} 1 \otimes (1+g) & (1-g) \otimes 1 & 0 & 0 & 0 \\ 0 & -1 \otimes (1-g) & (1+g) \otimes 1 & & \\ 0 & & 1 \otimes (1+g) & (1-g) \otimes 1 & \\ 0 & & & -1 \otimes (1-g) & (1+g) \otimes 1 \end{bmatrix} \\
 P_3 = \mathbb{Z}[C_2] & \xrightarrow{(1 \otimes 1) \quad -g \otimes g \quad -1 \otimes 1 \quad g \otimes g} & (P_0 \otimes P_3) \oplus (P_1 \otimes P_2) \oplus (P_2 \otimes P_1) \oplus (P_3 \otimes P_0) \\
 \downarrow 1-g & & \downarrow \begin{bmatrix} 1 \otimes (1-g) & (1-g) \otimes 1 & 0 & 0 \\ 0 & -1 \otimes (1+g) & (1+g) \otimes 1 & \\ 0 & & 1 \otimes (1-g) & (1-g) \otimes 1 \end{bmatrix} \\
 P_2 = \mathbb{Z}[C_2] & \xrightarrow{(1 \otimes 1) \quad g \otimes g \quad -1 \otimes 1} & (P_0 \otimes P_2) \oplus (P_1 \otimes P_1) \oplus (P_2 \otimes P_0) \\
 \downarrow 1+g & & \downarrow \begin{bmatrix} 1 \otimes (1+g) & (1-g) \otimes 1 & 0 \\ 0 & -1 \otimes (1-g) & (1+g) \otimes 1 \end{bmatrix} \\
 P_1 = \mathbb{Z}[C_2] & \xrightarrow{(1 \otimes 1) \quad -g \otimes g} & (P_0 \otimes P_1) \oplus (P_1 \otimes P_0) \\
 \downarrow 1-g & & \downarrow [1 \otimes (1-g) \quad (1-g) \otimes 1] \\
 P_0 = \mathbb{Z}[C_2] & \xrightarrow{(1 \otimes 1)} & P_0 \otimes P_0 = \mathbb{Z}[C_2] \otimes \mathbb{Z}[C_2] \\
 & \searrow & \swarrow \\
 & \mathbb{Z} &
 \end{array}$$

So, $x \cdot x$ is represented by

$$P_4 = \mathbb{Z}[C_2] \xrightarrow{-1 \otimes 1} P_2 \otimes P_2 \xrightarrow{f_x \otimes f_x} \mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}$$

This composition sends 1 to -1. So $x \cdot x$ is a generator of $H^4(C_2)$. Similarly, we can know x^n is a generator in $H^{2n}(C_2)$. So $H^*(C_2) \cong \mathbb{Z}[x]/(2x)$.